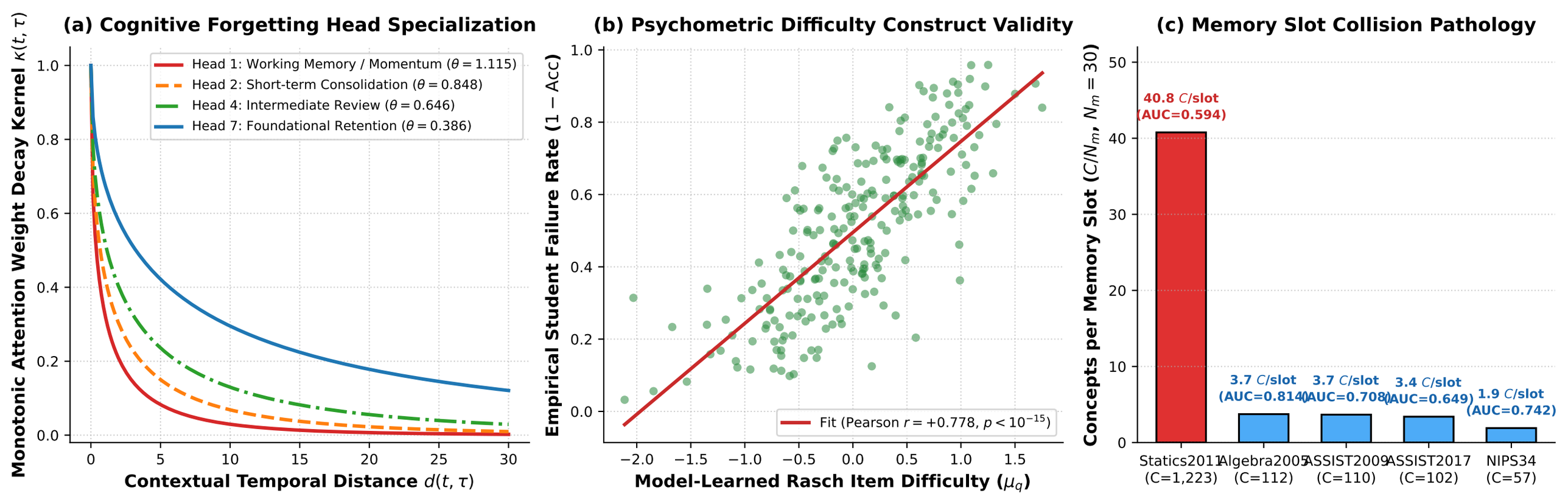
(a) Cognitive Forgetting Head Specialization
Monotonic Attention Weight Decay Kernel $\kappa(t,\tau)$
Contextual Temporal Distance $d(t,\tau)$
Head 1: Working Memory / Momentum ($\theta = 1.115$)
Head 2: Short-term Consolidation ($\theta = 0.848$)
Head 4: Intermediate Review ($\theta = 0.646$)
Head 7: Foundational Retention ($\theta = 0.386$)
(b) Psychometric Difficulty Construct Validity
Empirical Student Failure Rate $(1 - \text{Acc})$
Model-Learned Rasch Item Difficulty (μ_q)
Fit (Pearson $r = +0.778$, $p < 10^{-15}$)
(c) Memory Slot Collision Pathology
Concepts per Memory Slot (C/N_m, $N_m = 30$)
40.8 C/slot
(AUC=0.594)
3.7 C/slot
(AUC=0.814)
3.7 C/slot
(AUC=0.708)
3.4 C/slot
(AUC=0.649)
1.9 C/slot
(AUC=0.742)
Statics2011
(C=1,223)
Algebra2005
(C=112)
ASSIST2009
(C=110)
ASSIST2017
(C=102)
NIPS34
(C=57)